- Annual temperature varies from -30° C to 30° C.
- Precipitation (750-1500 mm) is distributed evenly throughout the year.
- Soil is fertile, enriched with decaying litter.
- Canopy is moderately dense and allows light to penetrate, resulting in well-developed and richly diversified understorey vegetation and stratification of animals.
- Flora is characterized by 3-4 tree species per km^2. Trees have broad leaves that are lost annually such as oak, hickory, beech, hemlock, maple, basswood, cottonwood, elm, willow, and spring-flowering herbs.
- Fauna consists of squirrels, rabbits, skunks, birds, deer, mountain lion, bobcat, timber wolf, fox, and black bear.

Only scattered remnants of original temperate forests remain today.

Desert biomes

- Deserts cover about one fifth of the earth's surface and occur where rainfall is >500 mm/year.
- Rainfall is usually very low and/or concentrated in short bursts between long rainless periods. Evaporation rates regularly exceed rainfall rates.
- Soils are course-textured, shallow, rocky or gravely with good drainage and have no subsurface water. The finer dust and sand particles are blown elsewhere, leaving heavier pieces behind. Sand dunes are common.
- Mean annual temperatures range from 20-25° C. The extreme maximum ranges from 43.5 - 49° C. Minimum temperatures sometimes drop to -18° C. Based on the temperature range, deserts can be Hot deserts and Cold deserts.
- **Hot deserts** such as the Sahara of North Africa and the deserts of the southwestern U.S., Mexico, Australia and India (Thar desert) occur at low latitudes.
- Hot deserts have a considerable amount of specialized vegetation (xerophytes), aloe, agave, Opuntia species, Euphorbia royleana as well as specialized vertebrate and invertebrate animals.
- The dominant animals of warm deserts are reptiles and small mammals. The Indian Spiny-tailed lizard, the blackbuck, the white-footed fox are the common fauna of the Thar deserts. There are also insects, arachnids and birds.
- **Cold deserts** occur in Antarctic, Greenland and the Nearctic realm, parts of USA and in parts of western Asia and the Ladakh region in India.
- Widely distributed animals are jack rabbits, kangaroo rats, kangaroo mice, pocket mice, grasshopper mice, antelope and ground squirrels.



Rainfall is lowest in the Atacama Desert of Chile, where it averages less than 15 mm. Some years are even rainless. Inland Sahara also receives less than 15 mm rainfall a year. Rainfall in American deserts is higher — almost 280 mm a year.

11.5 Responses To Abiotic Factors

Every living organism responds to its environment. There are various ways by which organisms respond to abiotic conditions. Some organisms can maintain constant physiological and morphological conditions or undertake steps to overcome the environmental condition, which in itself is a response (**Fig. 11.9).**

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